Set 1 **Reg No:**

CHRIST (Deemed to be University)
Department of Computer Science
Master of Computer Applications
Continuous Internal Assessment – I (CIA 1) Examination

Trimester: IV
Programme: MCA
Course Title: Machine Learning
Course Code: MCA 521-4
Course Type: Theory and Practical

Credits: 4
Total Marks: 30 Marks
Date :23/June/2026
Time duration: 2Hrs

All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.

- Verify the Course code / Course title & number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place in the hall.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer booklet. No sheets should be detached from the answer booklet.
- Answers without the question numbers indicated will not be valued. No page should be left blank in the middle of the answer booklet.

Course Outcome Mapped

CO1: Explain the fundamental principles and scope of Machine Learning
CO2: Implement modelling practices in machine learning for better generalisation
CO3: Interpret predictive modelling results and performance of supervised machine learning techniques
CO4: Assess the outcomes and effects of unsupervised machine learning processes
CO5: Deploy robust machine learning models for interdisciplinary applications

Part A Refute Question
Data Preprocessing and Machine Learning Debugging (CUT Method)

Time: 20 minutes **Max Marks: 10**

Instructions

Answer all questions.

Apply CUT method for each question:
C: Counter-example input
U: Expected correct output
T: Actual buggy output with trace

Show working clearly.

Q. No	Questions	CO/ RBT	Marks

1	The following function is supposed to replace missing values in a numerical column with the column mean.
	
	def fill_missing(col):
	    mean_val = sum(col) / len(col)
	    return [mean_val if x is None else x for x in col]
		(CO1, L4)	2M

2	The following function is supposed to scale values into the range [0,1].
	
	def min_max_norm(data):
	    mn = min(data)
	    mx = max(data)
	    return [(x - mn) / mx for x in data]
		(CO1, L4)	2M

3	The function is supposed to assign unique integers to each category.
	
	def label_encode(col):
	    mapping = {}
	    encoded = []
	    for i, val in enumerate(col):
	        mapping[val] = i
	        encoded. append(i)
	    return encoded
		(CO1, L4)	2M

4	The function is supposed to split data into 80% training and 20% testing sets.
	
	def train_test_split(data, ratio=0.8):
	    split = int(ratio * len(data))
	    return data[:split], data[split:]
		(CO1, L4)	2M

5	The function is supposed to compute Mean Squared Error.
	
	def mse (Y,):
	   n = len(Y)
	   return sum ((-) for i in range(n)) / n
		(CO1, L4)	2M

PART B   Lab Implementation

SDG Mapping**: **SDG Mapping: SDG 3 – Good Health and Well-being (Target 3.6: Reduce road traffic deaths and injuries)

This question is aligned with SDG 3: Good Health and Well-being, particularly Target 3.6, as it analyzes road accident patterns, severity, fatalities, injuries, emergency response, and economic loss to support safer mobility and accident prevention.

Time: 1:30hour   **Max Marks: 20**

Using the given Road Accident Dataset in GCR answers the following questions: -

Q. No	**Questions**	**CO/RBT**	**Marks**
1.	Load the given road accident dataset and display the first 10 rows. List all numerical and categorical variables separately. Check the data types of all columns and interpret whether any column needs type conversion	(L4, CO1)	2 M
2.	a. Generate descriptive statistics for all numerical variables. Which variable shows the highest variation? b. Compare the mean and median of Medical Cost. What does the difference indicate about Skewness?	(L3, CO1)	2 M
3.	a. Identify all variables containing missing values. Calculate the number and percentage of missing values in each column. b. Explain and identify which missing value technique can be used for variables Visibility Level, Traffic Volume, and Population Density. c. Explain and identify which missing value technique can be used for variables Driver Alcohol Level, Emergency Response Time, and Medical Cost. d. How can missingness in Driver Alcohol Level be both MAR and MNAR? Explain with reference to the dataset.	(L5, CO1)	4 M
4.	a. Plot a chart showing accidents by Weather Condition, Road Type and Accident Severity. b. Plot a boxplot for Economic Loss, Medical Cost grouped by Accident Severity. c. Plot a scatter plot between Traffic Volume and Accident Severity / Economic Loss.	(L3, CO1)	4 M

	d. Interpret any three charts and explain what they reveal about accident patterns.		
5.	Perform a correlation analysis on the numerical variables by creating a correlation matrix. Identify the strength and direction of the correlations, and interpret the relationships between Emergency Response Time, Medical Cost, Economic Loss, Driver Alcohol Level, and Accident Severity. Further, explain why categorical variables should not be directly included in correlation analysis without appropriate encoding.	(L4, CO1)	2 M
6.	a. Use the IQR method to identify outliers for Medical Cost, Economic Loss, Traffic Volume and Emergency Response Time. b. Examine whether high Medical Cost values are errors or meaningful extreme cases.	(L3, CO1)	2 M
7.	a. How do accident trends and impact metrics vary over time? b. How do environmental and road conditions contribute to accident outcomes? c. What factors most significantly influence accident severity and overall impact? d. How do accident characteristics and impacts vary geographically?	(L6, CO1)	4 M

Revised Bloom's Taxonomy (RBT) Levels:		
L1 – Remembering	**L2 – Understanding**	**L3 – Applying**
L4 – Analysing	**L5 – Evaluating**	**L6 – Creating**

Note: -

a. Students are required to complete the corresponding code for each scenario and submit their completed code as an ipynb file & PDF file.
b. It is expected that students provide comprehensive descriptions in their code (Syntax of the function used), specifying the purpose and usage of each function or package.